NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

SPMS25067- A regime-aware multi-agent deep reinforcement learning framework for financial problems

Presented by Menon Bharath Sreekumar

Supervised by Assoc Prof Patrick Chi Seng Pun

Introduction & Motivation

Problem: Conventional Reinforcement Learning (RL) agents cannot identify hidden regimes. Financial markets with their dynamic, probabilistic regime switching serve as an ideal testbed to address this limitation.

Research Hypothesis: *Agents equipped with explicit regime knowledge via a Mixture of Experts (MoE) architecture gated by a Hidden Markov Model (HMM) will demonstrate superior portfolio performance over regime-unaware baselines.*

Research objectives.

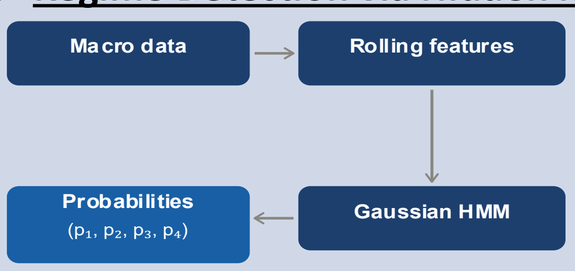
01 Propose a regime-aware MoE architecture gated by HMM probabilities.

02 Validate soft gating outperforms hard agent-switching at boundaries.

03 Demonstrate superior returns regardless of underlying RL algorithm.

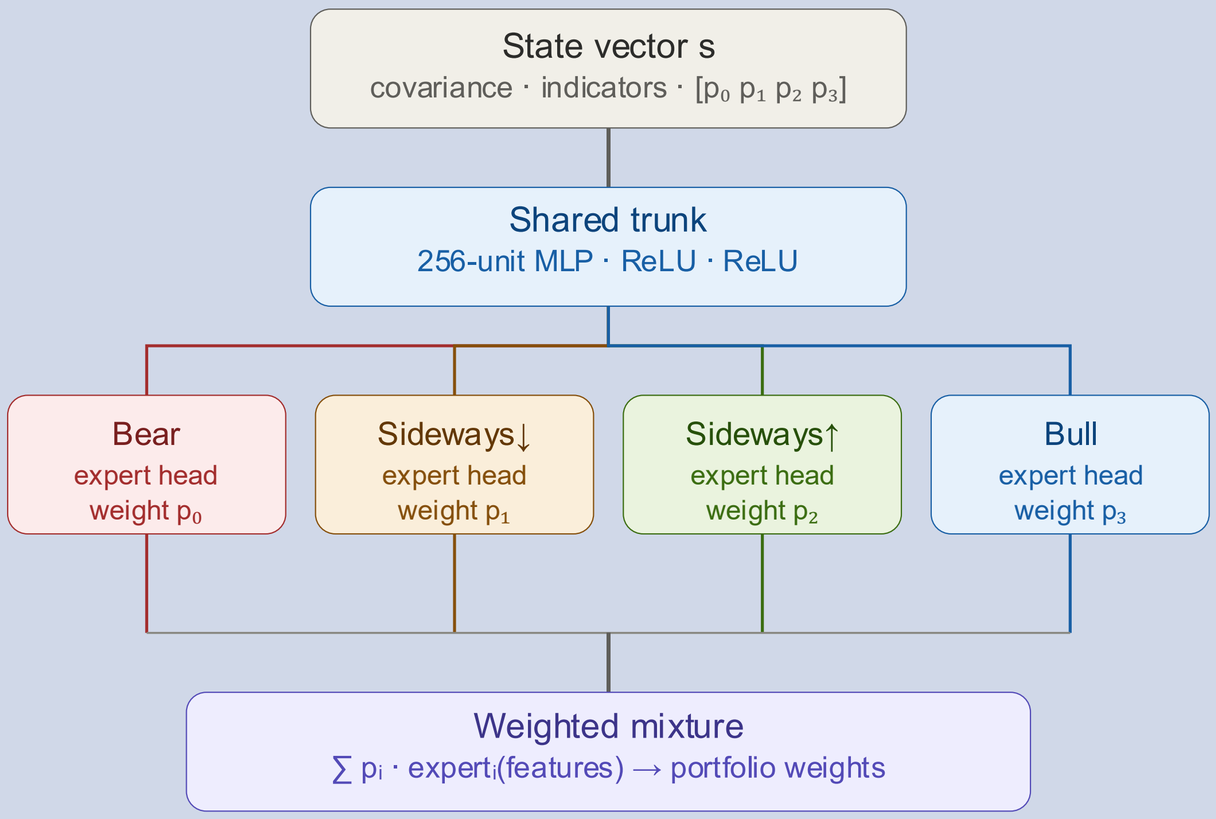
Methodology

1. Regime Detection via Hidden Markov Model



Features – Rolling volatility, Mean return, Assets' correlation.

2. Mixture of Experts Architecture



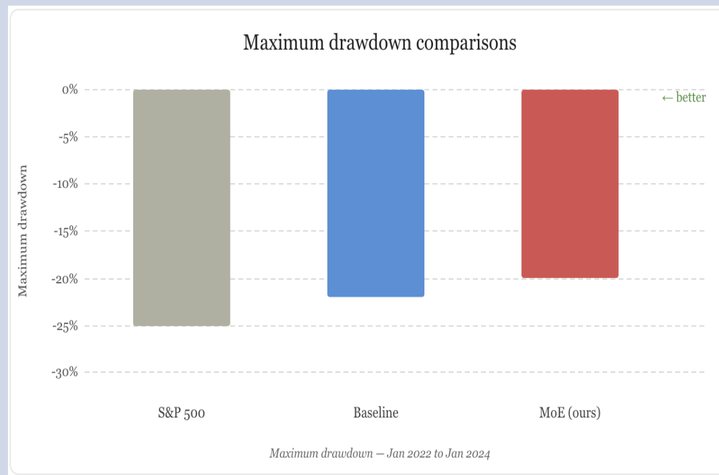
Reward Function

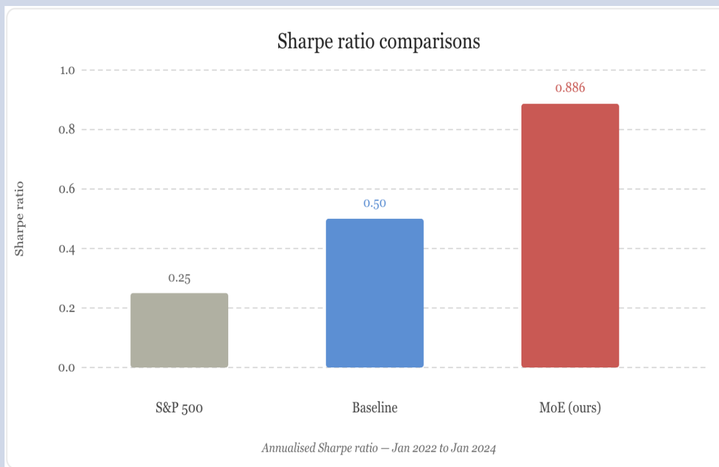
$$r_t = (R_t - \lambda \cdot w^T \Sigma w)$$

r_t — reward at timestep t
R_t — portfolio return at timestep t
λ — risk-aversion coefficient
w — portfolio weight vector
Σ — 20-day rolling covariance matrix

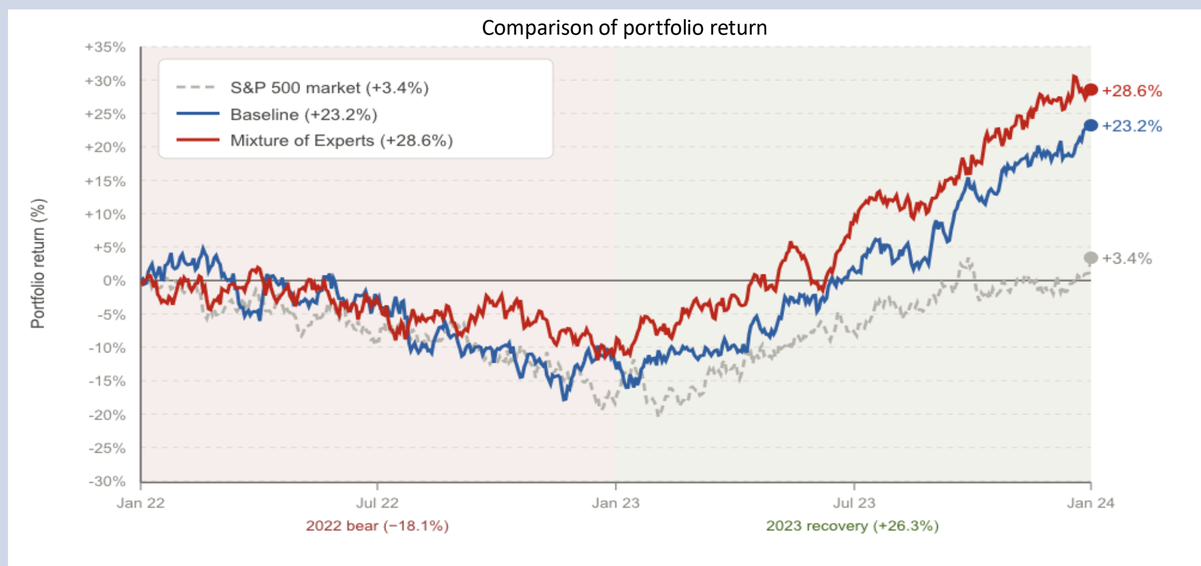
3.Evaluation Metrics: Out-of-sample evaluation uses **cumulative return**, **Sharpe ratio**, and **maximum drawdown** as metrics to compare against a single A2C baseline.

Results





As shown in the adjacent plots, the Mixture of Experts architecture achieved lower drawdown of 20% compared to the market's drawdown of 25% and baseline's 23%. On risk-adjusted returns, the MoE achieved a higher Sharpe ratio of 0.88 compared to 0.25 for the market and 0.5 for baseline. In all three of the evaluation metrics, the new architecture performed significantly better.



As shown above, the proposed framework achieves 28% return compared to the market return of 3% and baseline return of 23% in the period 2022-2024.

Why does the proposed framework achieve superior results?
The MoE outperforms by aligning with how markets actually behave. A single agent learns one averaged policy across all regimes, leading to a compromised strategy that's suboptimal everywhere. In contrast, the MoE trains specialized experts (e.g., bull vs bear) that don't interfere with each other.

Conclusion & Future Work

Conclusion: The regime-aware MoE-RL framework outperforms both the market benchmark and A2C baseline across all metrics (28% return, 0.88 Sharpe, 20% drawdown), validating that explicit HMM-gated regime knowledge yields more robust portfolio policies than conventional single-agent RL.

Future Work: Future work includes deriving a regime-conditioned policy gradient where the advantage function is weighted by HMM posteriors for tighter regime-policy integration. Replacing the HMM with a differentiable neural model would enable end-to-end training, and extension to multi-asset settings would test scalability.

References

Liu, X.-Y., Yang, H., Chen, Q., Zhang, R., Yang, L., Xiao, B., & Wang, C. D. (2020). FinRL: A Deep Reinforcement Learning Library for Automated Stock Trading in Quantitative Finance. *ArXiv:2011.09607 [Cs, Q-Fin]*. https://arxiv.org/abs/2011.09607

Istiaque, R. A., Pun, C. S., & Yong, B. Y. S. (2025). Stock market simulator using hidden Markov generative model and its application in risk measurement. *Quantitative Finance*, *25*(6), 873–893. https://doi.org/10.1080/14697688.2025.2511115